

MTH 154: Quantitative Reasoning Project.

Percent Application on Receipts.

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Instructor: Samuel Chukwuemeka

Date: 10/08/2025

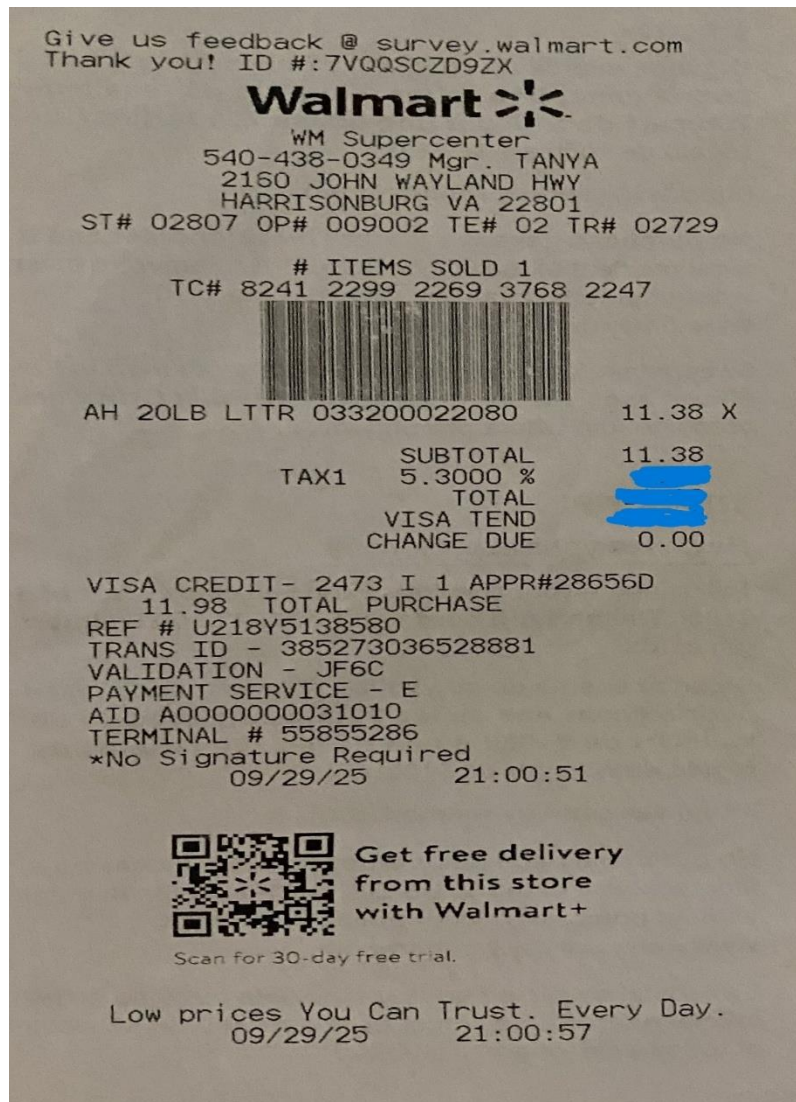
Objectives:

- (1.) Given the initial price and the tax rate, I will calculate the tax amount and total price. This is Case 1.
- (2.) Given the initial; price and the tax amount, I will calculate the tax rate. This is Case 2.
- (3.) Given the tax rate and the total price, I will calculate the initial price (subtotal) and tax amount. This is Case 3.
- (4.) Given initial price (subtotal), tax rate, tax amount, weight, I will calculate unit price per pound. This is Case 4.
- (5.) Given initial price (subtotal), amount of discount, tax rate, tax amount, sale price (Balance), I will calculate the exact percent discount. This is Case 5.

Case 1

Given initial price (subtotal), tax rate

To calculate tax amount and total price



$$\text{Tax rate} = 5.3000\% = \frac{5.3000}{100} = 0.053$$

$$\text{Subtotal} = \$11.38$$

$$\text{Tax} = \text{Tax rate} * \text{Subtotal}$$

$$\text{Tax} = 0.053(11.38)$$

$$\text{Tax} = 0.60314 \approx \$0.60$$

$$\text{Total Price} = \text{Subtotal} + \text{Tax}$$

$$\text{Total Price} = 11.38 + 0.60314$$

$$\text{Total Price} = 11.98314 \approx \$11.98$$



Case 2

Given initial price (subtotal), tax amount

To calculate tax rate



$$\textit{Subtotal} = \$ 50.97$$

$$\textit{Tax amount} = \$ 6.27$$

what % of \$ 50.97 is \$ 6.27

Percent – Equation

$$\frac{\textit{what}}{100} * 50.97 = 6.27$$

$$\textit{what} = \frac{6.27 * 100}{50.97}$$

$$\textit{what} = 12.30135314 \%$$



1311 S High St
Harrisonburg VA (22801)
(540) 209-1860

10/20/2025, 6:06 PM
Gabriela Z

Ref# 19931
Cashier: Gabriela Z

Para Llevar# 19931

2 x 33. Tlayuda	\$33.98
Pollo Chicken	\$0.00
Note:- Una no cevolla	
33. Tlayuda	\$16.99
Camitas Fried Pork	\$0.00
Note:- No cevolla	

SubTotal	\$50.97
Total Tax	\$6.27

GUEVARA/IVETTE

Visa
Chip
AID: A0000000031010
Non-Cash Adj
Total Due
Paid On VISA (3255)
Balance Due:

Auth: 10029D
Acc# 3255

\$2.28
\$59.52
\$59.52
\$0.00

Va Tax(5.3%)
Meal Tax(7%)
Thank You For Visiting Us!

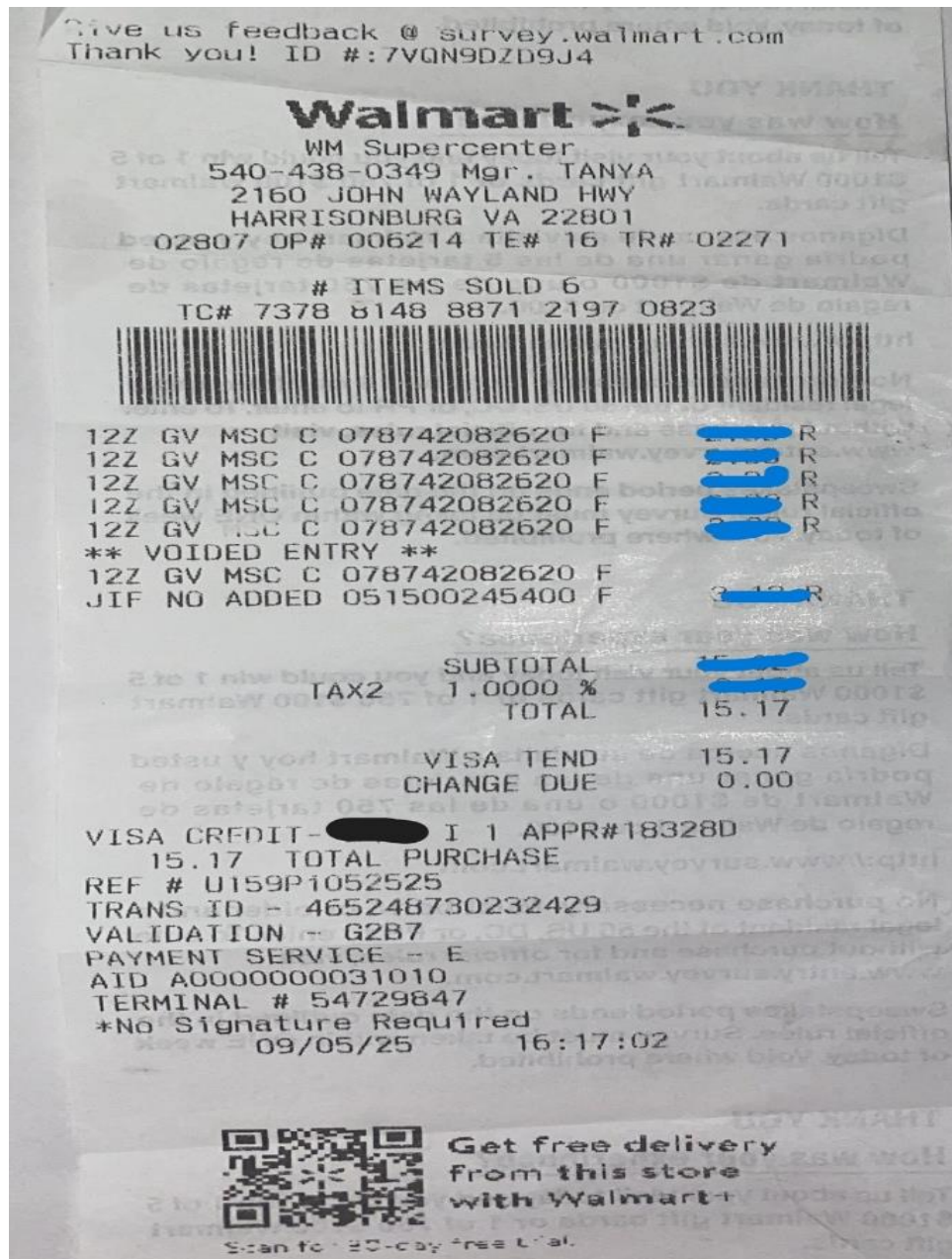
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Register 1
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Case 3

Given tax rate and the total price

To calculate initial price (subtotal) and tax amount



Let the subtotal (sum of initial price of the items) = p

$$a) 1.0000\% = \frac{1.0000}{100} = 0.01$$

$$\text{Subtotal} = p$$

$$\text{Tax} = 0.01(p) = 0.01p$$

$$\text{Total price} = p + 0.01p$$

$$\text{Total price} = 1.01p$$

$$1.01p = 15.17$$

$$p = \frac{15.17}{1.01}$$

$$p = 15.01980198$$

$$p \approx \$15.02$$

$$b) \text{Tax} = 0.01(15.01980198)$$

$$\text{Tax} = 0.1501980198$$

$$\text{Tax} \approx \$0.15$$

Give us feedback @ survey.walmart.com
Thank you! ID #:7VQN9DZD9J4

Walmart 

WM Supercenter
540-438-0349 Mgr. TANYA
2160 JOHN WAYLAND HWY
HARRISONBURG VA 22801
02807 OP# 006214 TE# 16 TR# 02271

ITEMS SOLD 6
TC# 7378 8148 8871 2197 0823



12Z	GV	MSC	C	078742082620	F	2.38	R
12Z	GV	MSC	C	078742082620	F	2.38	R
12Z	GV	MSC	C	078742082620	F	2.38	R
12Z	GV	MSC	C	078742082620	F	2.38	R
12Z	GV	MSC	C	078742082620	F	2.38	R
** VOIDED ENTRY **							
12Z	GV	MSC	C	078742082620	F		
JIF	NO	ADDED	051500245400	F		3.12	R

	SUBTOTAL	15.02
TAX2	1.0000 %	0.15
	TOTAL	15.17

	VISA TEND	15.17
	CHANGE DUE	0.00

VISA CREDIT-  I 1 APPR#18328D
15.17 TOTAL PURCHASE

REF # U159P1052525

TRANS ID - 465248730232429

VALIDATION - G2B7

PAYMENT SERVICE - E

AID A0000000031010

TERMINAL # 54729847

*No Signature Required

09/05/25 16:17:02



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Case 4

Given initial price (subtotal), tax rate, tax amount, weight

To calculate unit price per pound

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Thank you! ID #:7VRBDRZD7K1

Walmart *

WM Supercenter
540-438-0349 Mgr. TANYA
2160 JOHN WAYLAND HWY
HARRISONBURG VA 22801
ST# 02807 OP# 009001 TE# 01 TR# 00377

ITEMS SOLD 5
TC# 8103 1803 3228 9428 0126



CUCUMBER	000000040620 F	
3 AT 1 FOR	0.72	2.16 R
BANANAS	000000040110 F	
2.680 lb. @ 1 lb. /		R
LIME BULK	000000040480 F	
6 AT 1 FOR	0.25	1.50 R
TOMATO ROMA	000000040870 F	
1.200 lb. @ 1 lb. /	0.97	1.16 R
MANGO	000000040510 F	
3 AT 1 FOR	1.47	4.41 R
SUBTOTAL		10.68
TAX2	1.0000 %	0.11
TOTAL		10.79
VISA TEND		10.79
CHANGE DUE		0.00

VISA CREDIT- I 1 APPR#37244D
10.79 TOTAL PURCHASE
REF # 531209051476
TRANS ID - 465312721864509
VALIDATION - HL3J
PAYMENT SERVICE - E
AID A0000000031010
TERMINAL # 50865603
*No Signature Required
11/08/25 15:03:05



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11/08/25 15:03:12

second item

$$\text{Subtotal} = 10.68$$

Initial price of 2nd item

$$= 10.68 - 9.23$$

$$= \$1.45$$

Let unit price of 2nd *item* (*price per pound*) = k

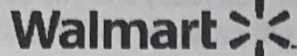
$$2.680 * k = 1.45$$

$$k = \frac{1.45}{2.680}$$

$$k = 0.5410447761$$

$$k \approx \$0.54$$

Give us feedback @ survey.walmart.com
Thank you! ID #:7VRBDRZD7K1



WM Supercenter
540-438-0349 Mgr. TANYA
2160 JOHN WAYLAND HWY
HARRISONBURG VA 22801
ST# 02807 OP# 009001 TE# 01 TR# 00377

ITEMS SOLD 5
TC# 8103 1803 3228 9428 0126



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6 AT 1 FOR	0.25	1.50 R
TOMATO ROMA	000000040870 F	
1.200 lb. @ 1 lb. /0.97		1.16 R
MANGO	000000040510 F	
3 AT 1 FOR	1.47	4.41 R

SUBTOTAL	10.68
TAX2 1.0000 %	0.11
TOTAL	10.79
VISA TEND	10.79
CHANGE DUE	0.00

VISA CREDIT- [REDACTED] I 1 APPR#37244D
10.79 TOTAL PURCHASE
REF # 531209051476
TRANS ID - 465312721864509
VALIDATION - HL3J
PAYMENT SERVICE - E
AID A0000000031010
TERMINAL # 50865603
*No Signature Required
11/08/25 15:03:05



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Case 5

Given initial price (subtotal), amount of discount, tax rate, tax amount, sale price
(Balance)

Calculate the exact percent discount.

FOOD LION
Food Lion #0266 (540) 433-8099
1751 South High St Harrisonburg, VA

DELI
COOKIE & CRM CPCK 6P 5.49 A *

GROCERY
FL MVI THIR BTR PCRN 1.49 A *
COKE CLASSIC 2L M 3.49 A *
1 @ 2 FOR 6.98
COKE CLASSIC 2L M 3.49 A *
1 @ 2 FOR 6.98

Savings
You saved:
COKE CLASSIC 2L -1.98
Shop&Earn SAVINGS! -10.00 *

Tax Paid
1.00% Tax 1 1.98 0.02
4 BALANCE DUE 2.00
VISA \$2.00

MID: 160000101711
RRN: 049809

SALE
XXXXXXXXXXXX2473 Entry Method: Chip
VISA 12:15:33
09/28/2025 049809
INVOICE:
VISA CREDIT
AID: A0000000031010
TVR: 8000008000
TSI: 6800
Total: USD\$ 2.00
APPROVED 68063D

CHANGE 0.00
You saved \$11.98!
* Year to Date
* Savings 157.33

We value your opinion!
Take our new survey
it only takes a few minutes.
Within the next 5 days please visit:
www.TalkToFoodLion.com
También disponible en español.
Use the PIN # below to start:
PIN: 0928026650040018

Loyalty Customer 46943959776
SHOP & EARN
You've redeemed your Shop & Earn
rewards on this transaction!
To see your remaining balance, visit
foodlion.com/rewards
STORE: 00266 REGISTER: 004 CASHIER: 0126
TICKET#: 0018 28SEP2025 12:15:36
www.foodlion.com
CUSTOMER SERVICE 1-800-210-9569
THANK YOU FOR SHOPPING AT FOOD LION

$$\text{Regular price} = \$13.96$$

$$\text{Sale Price} = \$1.98$$

$$\text{Amount of Discount} = \text{Regular price} - \text{Sale price}$$

$$\text{Amount of Discount} = 13.96 - 1.98$$

$$\text{Amount of Discount} = \$11.98$$

$$\text{Let \% discount} = P$$

$$P\% \text{ of regular price} = \text{Amount of Discount}$$

$$P\% \text{ of } 13.96 = 11.98$$

$$\frac{P}{100} * 13.96 = 11.98$$

$$\frac{100}{13.96} * \frac{P}{100} * 13.96 = \frac{100}{13.96} * 11.98$$

$$P = 85.81661891 \approx 85.82\%$$

The discount is approximately 85.82% rounded to two decimal places.

Reference (MLA9)

Chukwuemeka, Samuel. "*Percent Applications in Receipts.*" *Arithmetic/Algebra*

Project: Percents: Show Me Your Receipts,

[conferencepresentations.appspot.com/Projects/ArithmeticAlgebra/Receipts.h](https://conferencepresentations.appspot.com/Projects/ArithmeticAlgebra/Receipts.html)

tml. Accessed 20 Oct. 2025